No Current, No Braking: Two Instrument Lessons From a Magnet–Coil Damping Experiment

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Abstract

A magnet oscillating on a spring above a coil is a standard demonstration of electromagnetic braking, usually checked by showing that the damping scales as $1/R$ with the resistance of the coil circuit. We attempted this check with a US$43 data-acquisition system built around a Raspberry Pi Pico. Across series resistances from 15 Ω to 2.2 MΩ the decay rate did not change: $\gamma = (11.9 \pm 0.5) \times 10^{-3}\,\mathrm{s}^{-1}$ ($n = 8$). The coil was closed only through the amplifier's 1 MΩ bias resistor, so the braking current was about 16 nA and the electromagnetic damping a few millionths of the total. For spring 1, a loop closed through the coil alone would have added about 17%, which the data disfavour at 2.3–3 standard deviations, depending on the assumed amplitude. A second hidden error was the clock: mains hum recorded at 63 Hz instead of 50 Hz showed that each stored sample took 26% longer than the firmware assumed. The correction agrees with independent video timing and makes the frequencies consistent with the spring constants if two spring labels were interchanged. The remaining damping is mechanical and amplitude dependent. We give a corrected circuit and five checks for instructors.

Keywords: electromagnetic induction, damping, Lenz's law, data acquisition, null result, physics education

Introduction

When a magnet moves near a closed conducting loop, the changing flux drives a current whose magnetic field opposes the motion (Lenz's law). For a magnet oscillating on a spring above a coil, this current adds a velocity-proportional braking force, and the oscillation decays faster. Quantitative versions of this experiment, with the coil EMF logged by a computer interface, are well established (Kingman et al., 2002; Singh et al., 2002), and the related falling-magnet and eddy-current problems have been analysed in detail (Hahn et al., 1998; Saslow, 1987).

The usual way to show that the extra damping is electromagnetic is to change the resistance of the coil circuit. The dissipated power, and hence the electromagnetic damping, scales as $1/R_{\text{loop}}$. We attempted that measurement during a summer internship. No digital oscilloscope was available, so we built a low-cost data-acquisition system (DAQ) instead. The damping did not change with resistance.

This paper explains why. The explanation contains two lessons that apply to any coil experiment run through a microcontroller. First, measuring the voltage across a coil is not the same as letting current flow through it. Second, a microcontroller's sample rate cannot be assumed; it must be measured, and the mains hum already present in the data is a free way to do so. We report both, together with the mechanical damping that was actually measured and a corrected circuit.

Theory

Damped Oscillator

With a linear damping force $-b\dot{z}$, a mass m on a spring of constant k obeys $m\ddot{z} + b\dot{z} + kz = 0$. In the underdamped case, the solution is

$$z(t) = A_0 e^{-\gamma t}\cos(\omega t - \phi), \qquad \gamma = \frac{b}{2m}, \qquad \omega \approx \omega_0 = \sqrt{k/m}, \tag{1}$$

since in our data $\gamma/\omega \approx 10^{-3}$ (French, 1971). The quality factor is $Q = \omega/(2\gamma) = \pi f/\gamma$. Independent loss channels add, so $\gamma = \gamma_{\text{mech}} + \gamma_{\text{em}}$.

Electromagnetic Damping From the Measured EMF

If $\Phi(z)$ is the magnet's flux through one of the coil's N turns, Faraday's law gives

$$\varepsilon = -N\frac{d\Phi}{dz}\,\dot{z} \equiv -\kappa(z)\,\dot{z}. \tag{2}$$

For a dipole on the axis of a loop of radius a, $\Phi \propto a^2/(z^2+a^2)^{3/2}$, so $\kappa(z)$ is zero at the coil plane and largest at $z = \pm a/2$ (Griffiths, 2017). Because κ changes along the stroke, the EMF is not a pure sinusoid even when the motion is. It contains harmonics that grow with amplitude, as the recordings below show. This matters for the analysis below.

If the coil is closed through a total resistance R_{loop} (its inductance is negligible at 2 Hz), the Joule power averaged over a cycle is $\langle\varepsilon^2\rangle/R_{\text{loop}}$. Equating this to the rate of energy loss $2\gamma E$, with $E = \frac{1}{2}m\omega^2A^2$, gives

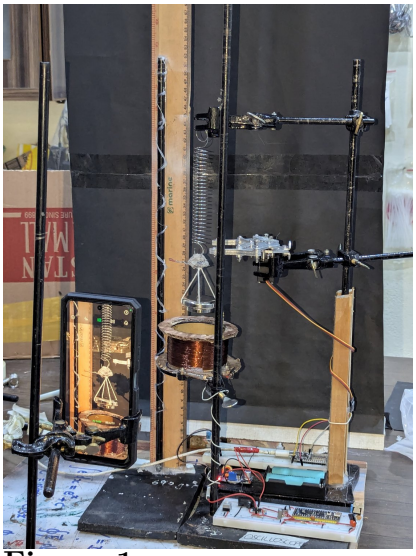
$$\gamma_{\text{em}} = \frac{\langle\varepsilon^2\rangle}{R_{\text{loop}}\,m\,\omega^2A^2} \equiv \frac{c}{R_{\text{loop}}}. \tag{3}$$

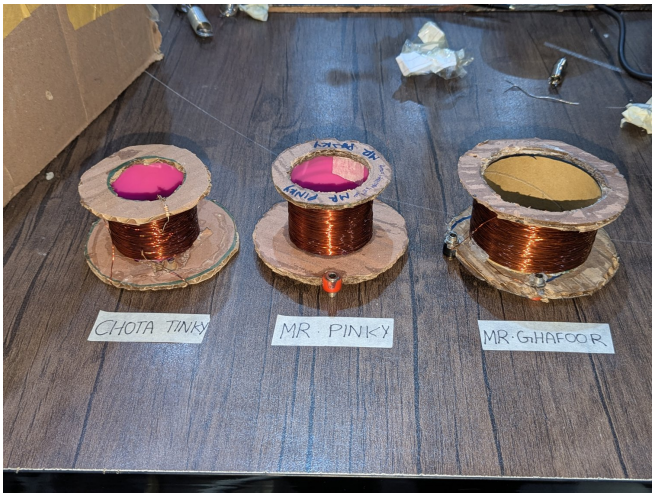
A high-impedance amplifier measures the open-circuit EMF, so Equation 3 predicts γ_{em} for any chosen R_{loop} from a single recording. It requires knowledge of neither N nor the magnet's dipole moment.

Method

Apparatus

Two NdFeB disc magnets, stuck together, hung in a string cradle from a helical spring (Figure 1). The total oscillating mass was 105 g. Each disc is about 4 cm in diameter and 0.7 cm thick, estimated from photographs against a metre rule; the grade was not recorded. A Hall probe gave 11 G at the coil's position. The magnet oscillated above a hand-wound coil of enamelled copper wire on a card former (coil 1, DC resistance $R_c = 15.2\,\Omega$, inner diameter about 5.6 cm, winding height about 4 cm). At rest, the magnet's centre was $z_0 = 6$ cm above the coil's centre. The turns were not counted. Treating the magnet as a dipole that gives 11 G at z_0, the measured EMF implies about 650 turns, which at this radius is about 120 m of wire; together with R_c this suggests wire





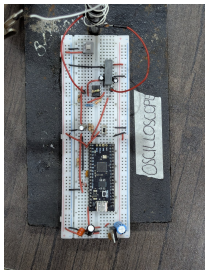


Figure 1

Apparatus

Note. Left: oscillator. The magnets hang in a string cradle from the spring, above coil 1; a servo claw releases the cradle, and a phone records the motion against a metre rule. Note the steel retort rod a few centimetres from the magnet. Top right: the three coils. Bottom right: the breadboard DAQ.

about 0.4 mm in diameter. None of these values enters the analysis, because Equation 3 uses the measured EMF directly. A servo-driven claw released the cradle from rest at a fixed height. Two springs were used. Their static spring constants, were 16.578 N/m and 19.438 N/m. Each was found from the slope of a force–extension graph for several hanging masses, with the extensions read twice, once by eye and once from a phone photograph.

Data Acquisition

The coil EMF is tens of millivolts and changes sign, so it cannot go straight into a microcontroller's 0–3.3 V analogue-to-digital converter (ADC). Figure 2 shows the signal chain. A buffered divider sets a 1.65 V reference, so negative EMF can be represented. The

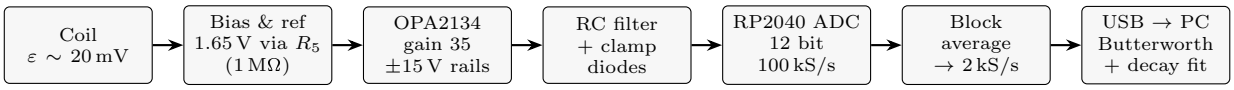


Figure 2

Signal Chain of the Data-Acquisition System

coil drives the high-impedance input of an OPA2134 amplifier (gain 35), whose input is biased to the reference through $R_5 = 1\,\mathrm{M\Omega}$. An RC filter (1.6 kHz) and clamp diodes protect the ADC of a Raspberry Pi Pico (Raspberry Pi Ltd, 2021). The firmware reads the ADC in a loop, averages blocks of 50 readings, stores up to 60 000 points in RAM (about 38 s at the true spacing) and sends the record over USB after each run. The parts cost US$43, and the design files, firmware and data are open (Malik et al., 2026).

The DAQ has several limitations, which we found during this analysis:

- The block average uses integer division, so each stored value is truncated to one ADC step. The averaging lowers noise but adds no resolution.
- The recorded step is 23.65 µV at the input, 2.7% larger than the repository's firmware constants give. This means the flashed firmware differed slightly from the published version. All EMF values, and hence c, carry this 2.7% scale uncertainty.
- The input-referred noise before each release ranged from 29 µV to 88 µV rms (median 48 µV). That is about 3×10^{-3} of the signal.
- The ADC timestamps are not measured. They are computed as sample number × 500 µs. This turned out to be wrong (Lesson 2 below).
- The op-amp runs from ±15 V rails. Under overload, the clamps can let the ADC pin exceed its absolute ratings. A single-supply rail-to-rail amplifier with Schottky clamps is safer (Horowitz & Hill, 2015).

Protocol

With coil 1, an external resistor R_{ext} was placed in series with the coil. It was set to 0, 33.6, 81.4, 324, 1051, 3180 and 9610 Ω for spring 1, with 0 Ω repeated, and to similar values for spring 2. One spring-1 run used 2.2 MΩ. Coils 2 and 3 were recorded once on each spring. Each run lasted about 38 s, or about 70–80 cycles. A phone filmed every run for video tracking.

Analysis

The analysis scripts are in the repository. Each record was low-pass filtered (fourth-order zero-phase Butterworth, 20 Hz; Virtanen et al., 2020). Because the harmonics in ε decay faster than the motion does, fitting the EMF peaks overestimates γ by about 20%. Instead, we isolated the fundamental with a band-pass filter at 0.6–1.4 f, took its Hilbert envelope (a smooth curve through the oscillation's peaks), and fitted ln(envelope) against time. The fit started about 1.3 s after the first large peak and ended about 1.3 s before the end of the record, with one point per cycle. Uncertainties come from a block bootstrap, which refits many resampled versions of each record. Two other estimators of the displacement decay (a nonlinear fit of the fundamental, and a model of the transduction) agree within 7% (spring 1) and 12% (spring 2), and we quote their spread as a method uncertainty. Figure 3 shows a typical recording and its envelope.

Results

The Damping Ignores the Resistance

Figure 4 shows γ against $R_c + R_{\mathrm{ext}}$. Neither spring shows a trend. Over the eight spring-1 runs below 10 kΩ,

$$\gamma_1 = (11.9 \pm 0.5_{\mathrm{scatter}} \pm 0.4_{\mathrm{method}} \pm 0.12_{\mathrm{clock}}) \times 10^{-3}\ \mathrm{s}^{-1}, \qquad Q_1 = 551 \pm 24.$$

For spring 2, $\gamma_2 = (13.5 \pm 1.0) \times 10^{-3}\,\mathrm{s}^{-1}$ and $Q_2 = 436 \pm 32$. The 2.2 MΩ run, where the circuit is effectively open, gives $\gamma = 11.5 \times 10^{-3}\,\mathrm{s}^{-1}$, the same as the rest. These values use the corrected timebase described below.

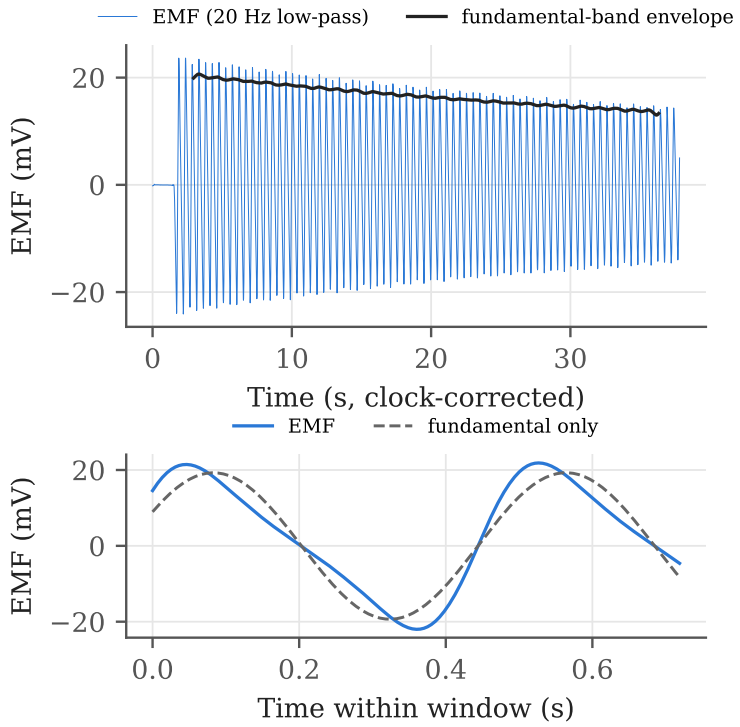


Figure 3

A Typical Recording (Spring 1, Coil 1, $R_{\text{ext}} = 0$)

Note. Top: EMF with the envelope of its fundamental. Bottom: the waveform is visibly non-sinusoidal because the coupling $\kappa(z)$ varies along the stroke.

From the measured EMF ($\varepsilon_{\text{rms}} \approx 16\,\text{mV}$ over the first three cycles) and the initial amplitude $A \approx 2.2\,\text{cm}$ seen in the video, Equation 3 gives $c = 0.030\,\Omega\,\text{s}^{-1}$. Closed through the coil alone, the loop would therefore have added $\gamma_{\text{em}} \approx 2.0 \times 10^{-3}\,\text{s}^{-1}$, about 17% of the total (9–25% for A between 3.0 and 1.8 cm). That effect is several times larger than the run-to-run scatter. A fit of $\gamma = \gamma_0 + c/(R_c + R_{\text{ext}})$ to spring 1 gives $c = (-0.011 \pm 0.006)\,\Omega\,\text{s}^{-1}$, which disfavours the predicted value by about 3 standard deviations if A is taken as exact, and by 2.3 if A is uncertain by 20%. A permutation test, which reshuffles the decay rates among the resistances, gives $p = .003$. The fitted c is itself slightly negative, which no physical loss could produce, so the run-to-run scatter is the limiting factor. The decisive argument is the circuit analysis below, not this fit.

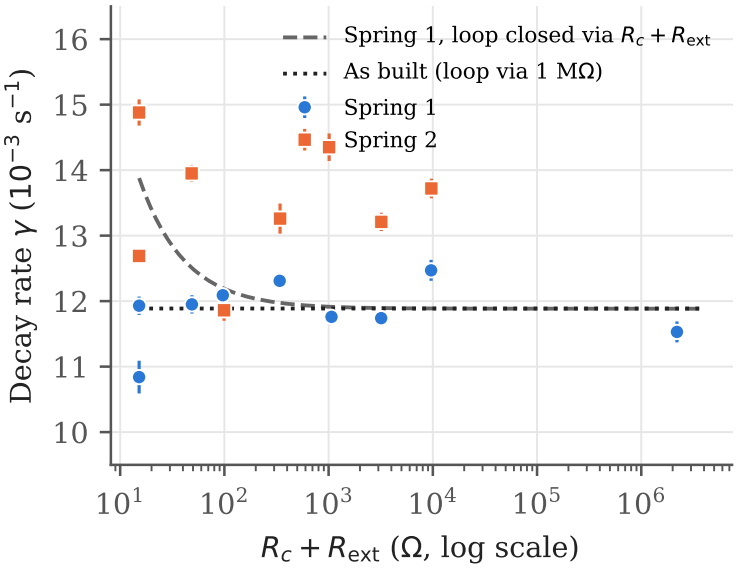


Figure 4

Decay Rate Against Coil-Side Series Resistance

Note. The dashed curve is Equation 3 for spring 1 if the loop had been closed through $R_c + R_{ext}$. The dotted line is the prediction for the circuit as built. Error bars are bootstrap standard errors.

Lesson 1: The Loop Was Never Closed

The coil connects between the 1.65 V reference and the non-inverting input of the amplifier (Figure 5a). The only path that closes the coil circuit is R_5. Adding R_{ext} in series changed R_{loop} from 1.000 MΩ to at most 1.010 MΩ. The current was always about 16 mV/1 MΩ ≈ 16 nA, dissipating about 0.3 nW. The mechanical loss is about 0.1 mW. By Equation 3, $\gamma_{em} \approx 3 \times 10^{-8}\,\mathrm{s}^{-1}$ in every run. The DAQ measured the EMF accurately, but no braking current ever flowed.

Two observations support this diagnosis.

- **The 2.2 MΩ run.** Here R_{ext} and R_5 form a voltage divider, so the recorded EMF should fall to $10^6/(3.2 \times 10^6) = 0.31$ of its usual value. After correcting for this run's 15% smaller release, it is 0.30.
- **The low resistances.** Small resistors made no measurable difference to the recorded EMF. That confirms they were wired in series with the coil: a 33.6 Ω resistor across

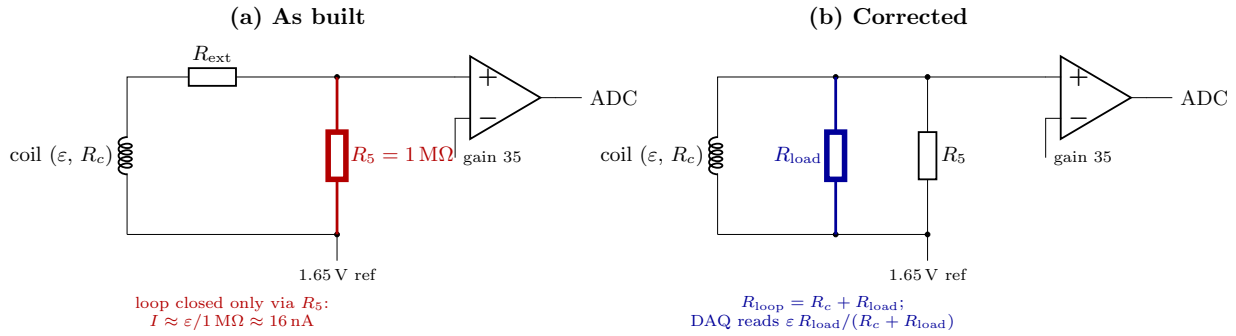


Figure 5

The Circuit as Built and a Corrected Version

Note. (a) The coil is closed only through the 1 MΩ bias resistor, so almost no current flows. (b) A load resistor across the coil terminals sets $R_{\mathrm{loop}} = R_c + R_{\mathrm{load}}$. The DAQ then reads $\varepsilon R_{\mathrm{load}}/(R_c + R_{\mathrm{load}})$.

the coil would instead have cut the recorded EMF to about 0.69 of its usual value.

The 2.2 MΩ run also shows the strongest mains pickup, as expected for a high-impedance node.

A corrected circuit. The braking current needs a known, low-resistance path. The simplest fix is to put R_{load} directly across the coil, in parallel with the amplifier input (Figure 5b), and to vary it. Comparing an open coil with a shorted coil gives γ_{em} directly. A transimpedance input does better: the coil and R_{ext} feed the inverting input of an op-amp with a feedback resistor. That closes the loop through only $R_c + R_{\mathrm{ext}}$, measures the current directly, and lets the original series-resistor protocol work unchanged.

Lesson 2: The Clock Was Never Checked

Every recording contains a narrow line at (63.07 ± 0.23) Hz, with a harmonic at 126.1 Hz (Figure 6). Pakistan's mains frequency is 50 Hz. If this line is mains pickup, the true spacing between stored samples is $s = 63.07/50 = 1.261 \pm 0.013$ times the spacing the firmware assumed; we call s the clock factor, and its uncertainty is set by the grid's tolerance. Three features identify the line as mains: it has harmonics, it is strongest in the high-impedance 2.2 MΩ run, and its frequency wanders by 0.4% between runs while the

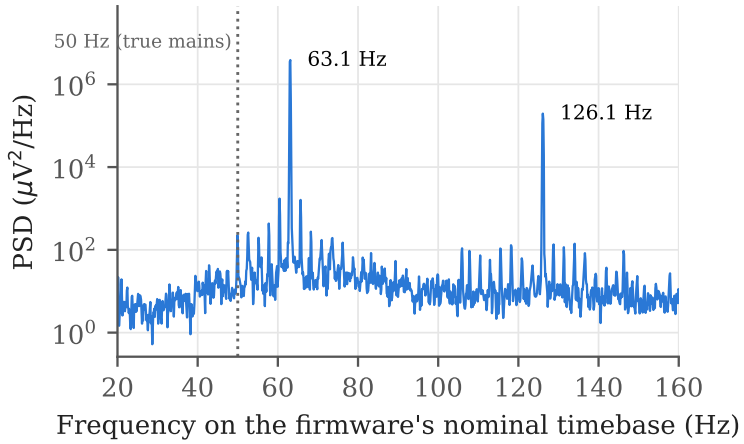


Figure 6

Mains Hum as a Clock

Note. Noise spectrum of the 2.2 MΩ run, plotted on the firmware's assumed timebase. The mains line, truly at 50 Hz, appears at 63.1 Hz.

spring frequency stays fixed to 0.02%, as expected for the grid and not for a crystal-timed source such as the servo controller. Lighting flicker would appear at 100 Hz. The cause is in the firmware. It schedules each ADC read 10 µs after the last. When one read plus the loop overhead takes longer, the loop simply runs flat out. Each read then takes about 12.6 µs, and each stored point is 631 µs apart, not 500 µs. A 60 000-point record therefore lasts 37.9 s, not 30 s; timing one recording with a stopwatch or video would confirm this directly.

The correction divides every frequency and decay rate by s, while leaving Q unchanged. Two independent checks agree with it:

- **Video.** The phone video runs on its own clock. The ratio of its decay rates to the DAQ values is 1.06 ± 0.06 with the clock correction and 0.84 ± 0.05 without it, so the video favours the correction. Because the per-run video rates scatter widely, this checks the overall scale only; counting oscillation cycles in the video would be a much tighter test.

- **Spring constants.** The corrected frequencies are 2.08 Hz and 1.86 Hz. Their ratio, 1.118, does not depend on s. It is consistent with the measured spring constants only

if the two k labels were interchanged (predicted 1.083). With the labels swapped, both frequencies lie 4–7% below the static-k predictions. Spring mass lowers the frequency by only about 1%, and because k came from the slope of a force–extension graph, initial tension in the spring does not bias it. The remaining gap is unexplained; an error in the oscillating mass or in the k measurements is possible. Re-measuring k, and timing the DAQ directly, would settle both points; we could not repeat the measurements after the internship.

The Damping That Remains Is Mechanical

Since γ_{em} was negligible, the measured damping is entirely mechanical, and it is not exponential. The first half of each run decays faster than the second, by $11 \pm 12\%$ for spring 1 (7 of 8 runs) and $19 \pm 6\%$ for spring 2 (all 8 runs). For spring 2, whose release amplitude varied more, γ also correlates strongly with the initial EMF ($r = .97$, Figure 7); for spring 1, whose release amplitude varied by only 6%, there is no such trend ($r = -.40$). The amplitude dependence is therefore clear for spring 2 and weaker for spring 1. Air drag on the cradle is about 2% of the damping, too small to explain this. Candidates are coupling to the cradle's pendulum or torsional modes (Berg & Marshall, 1991), friction at the string attachments, and losses in the steel retort rod a few centimetres from the magnet. We cannot yet tell them apart. A control run with the rod moved away would test the last.

The coil geometry changes the EMF but, with no current flowing, not the damping. Coil 2, with a different former diameter, gave 0.37 (spring 1) and 0.45 (spring 2) of coil 1's EMF. Coil 3, with a different wire gauge, gave about 0.11. The decay rates for coils 2 and 3 were the same as for coil 1 within the scatter.

Discussion

Neither error is exotic. A high-impedance input is the right choice for measuring an EMF and the wrong one for producing braking. A microcontroller loop that cannot keep up with its schedule fails silently. Both errors left clean-looking data: smooth exponential decays and a stable frequency. The first was revealed only because the expected $1/R$

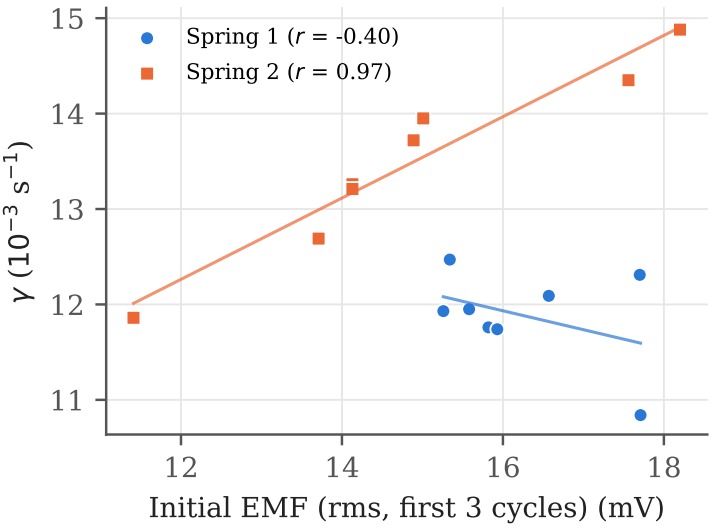


Figure 7

Decay Rate Against Initial EMF, a Proxy for Release Amplitude

dependence was missing. The second was revealed by the mains line, which we went looking for because the measured frequencies disagreed with the spring constants. The checks below would have caught both in the first hour.

Lessons for Instructors

1. Before blaming the physics, compute the current your circuit actually allows, $I = \varepsilon / R_{\text{loop}}$.
2. A high-impedance input on a coil measures the EMF but brakes nothing; give the braking current a known low-resistance path.
3. Do not trust a microcontroller's assumed sample rate. The 50 Hz (or 60 Hz) hum already in your data is a free clock.
4. Before collecting a full data set, run an open-coil versus shorted-coil control and check that the predicted effect exceeds your run-to-run scatter.
5. Compare the early and late halves of each decay; a single exponential fit can hide amplitude-dependent losses.

Our study has limitations. Each resistance was recorded once (twice at 0 Ω). The coil and magnet geometry were not fully documented. The amplitude A in Equation 3 comes from video. The clock correction rests on the mains frequency rather than a direct timing measurement. None of these affects the central conclusion: the 1 MΩ path cut the current by a factor of $R_5/R_c \approx 7 \times 10^4$ compared with a coil closed on itself.

Conclusion

A low-cost DAQ recorded millivolt coil EMFs cleanly, yet changing the series resistance from 15 Ω to 2.2 MΩ did not change the damping of a magnet–spring oscillator. The coil circuit was closed only through a 1 MΩ bias resistor, so the electromagnetic damping was a few millionths of the total. Separately, mains hum showed that the DAQ's sample spacing was 26% longer than assumed; the correction agrees with video timing and, if two spring labels were swapped, with the spring constants. Both are easy mistakes, easy to detect, and worth teaching.

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